MILITARY SPECIFICATION
CONNECTORS, ELECTRICAL, MINIATURE, QUICK DISCONNECT
(FOR WEAPONS SYSTEMS)
ESTABLISHED RELIABILITY
(This specification is mandatory for use by
The Munition Command, Department of The Army)

1. SCOPE

1.1 This specification covers two Varieties of miniature, quick disconnect bayonet lock, connectors. The two Series within each Variety are not to be interchangeable or intermatable. Each Series within a Variety is both interchangeable and intermatable with the corresponding Series of the remaining Variety.

1.2 Description

1.2.1 Variety A - Crimp Termination

Both Series shall employ removable pin and socket contacts intended for crimp termination. Both Series shall be designed to assure proper orientation of the mating halves prior to electrical circuit closure. Both Series shall include hermetically sealed receptacles with fixed contacts for solder termination. In addition, one Series shall be designed so that electrical continuity is maintained between mating shells prior to establishment of conductor circuits by mating of pin and socket contacts, and in the reverse order during the disconnecting operation and to have the contacts so located as to be protected from handling damage and inadvertent electrical contact. The other Series shall be designed to provide low silhouette for minimum size and weight and include connectors with positive shell-to-shell grounding when mated.

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1.2.2 Variety B - Solder Termination

Both Series shall employ fixed pin and socket contact intended for solder termination, set in rigid dielectric, filled epoxy or vitreous, depending on the Class. Both Series shall be designed to assure power orientation of the mating halves prior to electrical circuit closure. In addition, one Series shall be designed so that electrical continuity is maintained between mating shells prior to establishment of conductor circuits by mating of pin and socket contacts, and in the reverse order during disconnecting operation, thereby providing protection against EMR; and to have the contacts so located as to be protected from handling damage and inadvertent electrical contact. The basic construction is the same for both Series. The other Series shall be designed to provide low silhouette for minimum size and weight.

1.3 Classification

Connectors fabricated to this specification shall be classified as follows:

1.3.1 Variety A - Crimp Termination

a. Series I - Grounded Scoop-Proof - 150°C (302°F) max. operating temp.

II - A. Low Silhouette - 150°C (302°F) max. operating temp.

II - B. Low Silhouette, High Temperature - 200°C (392°F) max. operating temp.

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b. Types:

(1) Plugs

(a) Straight

(b) Straight, Grounded

(c) Line

(2) Receptacles

(a) Box Mounting

(b) Wall Mounting

(c) Jam-nut Mounting

(d) Solder Mounting

c. Classes

(1) E - Environment resisting

(2) P - For potting - includes potting form

(3) T - Environment resisting but without rear accessories

(4) H - Hermetically sealed

d. Finishes

(1) A - Bright Cadmium Plate

(2) B - Olive-drab Cadmium Plate

(3) C - Anodic

(4) D - Fused Tin Plate

(5) E - Stainless Steel, Unplated

e. Styles

(1) P - Pin contact

(2) S - Socket contact

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f. Kinds

(1) N - Normal contact length

(2) PC - Printed circuit contact length

1.3.2 Variety B - Solder Termination

a. Series I - Grounded Scoop-Proof 177°C (351°F)

II - Low Silhouette - 150°C (302°F) maximum operating temp.

b. Finish

(1) Bright Cadmium

(2) Olive Drab

c. Types

(1) Plugs

(a) Straight

(b) Line

(2) Receptacles

(a) Box Mounting

(b) Wall Mounting

(c) Jam-nut Mounting

(d) Solder Mounting

d. Classes

(1) T - General Duty - pressurized (receptacles only)

(2) P - For potting - 20029P is pressurized

(3) H - Hermetically Sealed (Vitreous Insert)

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e. Styles

(1) P - Pin Contact

(2) S - Socket contact

f. Kinds

(1) N - Normal contact length

(2) PC - Printed circuit contact length

2. APPLICABLE DOCUMENTS

2.1 The following documents form part of this specification to the extent specified herein. Except where otherwise specified, the issued current at the date of release of this revision of this specification shall apply but, in the case of conflict, this specification shall govern.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Fiber
QQ-P-416	Plating, Cadmium, Electrodeposited
QQ-S-365	General Requirements For

Military

MIL-P-116	Preservation, Methods of
MIL-J-5624	Jet Fuel, Grades JP-4 and JP-5
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-L-9236	Lubricating Oil, Aircraft Turbine Engine, 400°F
MIL-T-10727	Tin Plating, Electrodeposited or Hot-dipped for Ferrous and Nonferrous Metals

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MIL-W-16878	Wire, Electrical, Type E, 200°C and 260°C 600 Volts (Insulated, High Temperature) (Navy)
MIL-B-22020	Bags, Transparent, Flexible, Heat Sealable, Volatile, Corrosion Inhibitor Treated
MIL-B-22205	Bags, Transparent, Flexible, Heat Sealable, for Packing Applications
MIL-W-22759	Wire, Electric, Fluorocarbon, Insulated Copper
MIL-G-45204	Gold Plating (Electrodeposited)
MIL-G-3545	Grease, Aircraft, High Temperature
MIL-H-5606	Hydraulic Fluid

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts by Variables for Percent Defective
MIL-STD-9	Screw Thread Conventions and Method of Specifying
MIL-STD-414	Sampling Procedures and Tables for Inspection
MIL-STD-790	Reliability Assurance Program for Electronic Parts Specifications
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Tests

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DRAWINGS

Military Standard Drawings

MS18113	Wire, Electric, Extruded TFE-Fluorocarbon-insulated, silver-coated, copper, 1,000 volt
MS21985	Wire, Electric, Extruded TFE-Fluorocarbon-insulated, silver-coated, copper, 600 volt
MS3191	Tool, Hand Crimp, Class I - for Contacts of Electric Connectors
MS3198	Tool, Hand Crimp, Miniature, Class I - for Electric Connector Contacts
MS27495	Tool, Contact, Connector, Assembly and Disassembly

PUBLICATIONS

Handbooks

H-28	Screw Thread Standards for Federal Services

(Copies of this document required by contractors in connection with specific procurement functions may be obtained from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20325)

AMS3325	Silicone Rubber, Fuel and Oil Resistant, 55-65
AMS3326	Silicone Rubber, Fuel and Oil Resistant, 50-65

(Copies of these documents required by contractors in connection with specific procurement functions may be obtained from the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, New York 10017.)

3. REQUIREMENTS

3.1 Qualification. Both connector Varieties furnished under this specification shall be products that have been tested and passed the qualification tests specified herein and have been listed or approved for listing on the applicable Qualified Products List.

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3.2 Specification Sheets. The individual part (connector) requirements shall be as specified herein and in accordance with applicable specification sheets. In the event of conflict between the specification sheet and this specification, the specification sheet shall govern.

3.3 Materials. Materials shall be as specified herein, however, when a definite material is not specified, a material shall be used which will enable the connectors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a quaranty for acceptance of the finished product.

3.3.1 Dissimilar metals. When dissimilar metals are employed in intimate contact with each other, suitable protection against electrolytic corrosion shall be provided as specified in requirement 16 of MIL-STD-454.

3.3.2 Components. Materials for specific components of the connector shall be as follows:

3.3.2.1 Variety A

3.3.2.1.1 Classes E, P, and T

a. Shell-Impact extruded or machined aluminum alloy

b. Coupling Ring, Jam Nut and Potting Ring-machined aluminum alloy

c. Insert-Reinforced epoxy resin (Mesa Plastic BX1288 or equivalent), color optional

d. Contacts

(1) Style P - Copper alloy

(2) Style S - Copper alloy. Hood or sleeve shall be corrosion-resistant steel

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e. Bayonet Pins-Corrosion-resistant steel

f. Grounding Spring-heat treated beryllium copper or corrosion-resistant steel

g. Filler Compound

RTV Silicone or equivalent

h. Gaskets, Grommet, and Interface Seals-silicone rubber

3.3.2.1.2 Class H

a. Shell

(1) Flange and Jam-nut Mounting-Machined Corrosion Resistant Steel

(2) Flange, Jam-nut and Solder Mounting-Machined Carbon Steel

b. Insert

(1) Fused vitreous material for Style P

(2) Vitreous seal and epoxy resin for Style S

c. Contacts

(1) Style P - Nicket-iron B52 alloy or equivalent

(2) Style S -

d. Bayonet Pins-Corrosion resistant steel

e. Gaskets and Interface Seals-Silicone Rubber

3.3.2.2 Variety B

3.3.2.2.1 Classes P and T

a. Shell - Impact extruded or machined aluminum alloy

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b. Coupling Ring, Hex Jam nut and Potting Ring - Machines aluminum Alloy

c. Insert - Reinforced epoxy resin (Mesa Plastic BX 1288 or equivalent), color optional

d. Contacts

(1) Style P - Copper alloy

(2) Style S - Beryllium Copper. Hood or sleeve shall be corrosion resistant steel

e. Bayonet Pins - Corrosion resistant steel

f. Grounding Spring - Heat treated beryllium copper or corrosion resistant steel

g. Filler Compound (Class T, Series I and II; Class P, Series I) RTV silicone or equivalent

h. Gaskets - Fluorosilicone rubber per ASM 3325 or 3326

3.3.2.2.2 Class H

a. Shell

(1) Flange and Jam-nut Mounting - Machined Corrosion Resistant Steel

(2) Solder Mounting - Machined Carbon Steel

b. Insert

(1) Fused vitreous material for pin contacts

(2) Vitreous seal and epoxy resin for socket contacts

c. Contacts

(1) Style P - Nickel-iron B52 alloy or equivalent

(2) Style S - Nickel-iron B52 alloy or equivalent

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d. Bayonet Pins - Corrosion resistant steel

e. Gaskets - Fluorosilicone rubber per AMS 3325 or 3326

3.4 Construction

3.4.1 Contacts

3.4.1.1 Variety A

Contacts shall be designed for crimp termination in all but Class H assemblies which shall be solderable. Dimensions for solder-type contacts (and Normal Kind) shall conform to figure 1. Crimp-type contacts (and PC Kind) shall conform to the applicable specification sheets.

3.4.1.1.1 Contact Arrangement. Contact arrangement shall be as shown on applicable specification sheet. Except for Class Y assemblies, the engaging end of the pin contacts in assembled connectors shall be located within .024-inch diameter of true position and the engaging end of socket contacts in assembled connectors shall be located within .015-inch diameter of true position.

3.4.1.2 Variety B - Contacts shall be molded (Classes T and P) or fused (Class H) into the inserts. Contact configuration, dimensions, and surface finish shall be as shown in figure 3, except for PC Kind which shall be as shown on applicable specification sheets.

3.4.1.2.1 Contact Arrangement. Contact arrangement shall be as shown on applicable specification sheet.

3.4.2 Inserts

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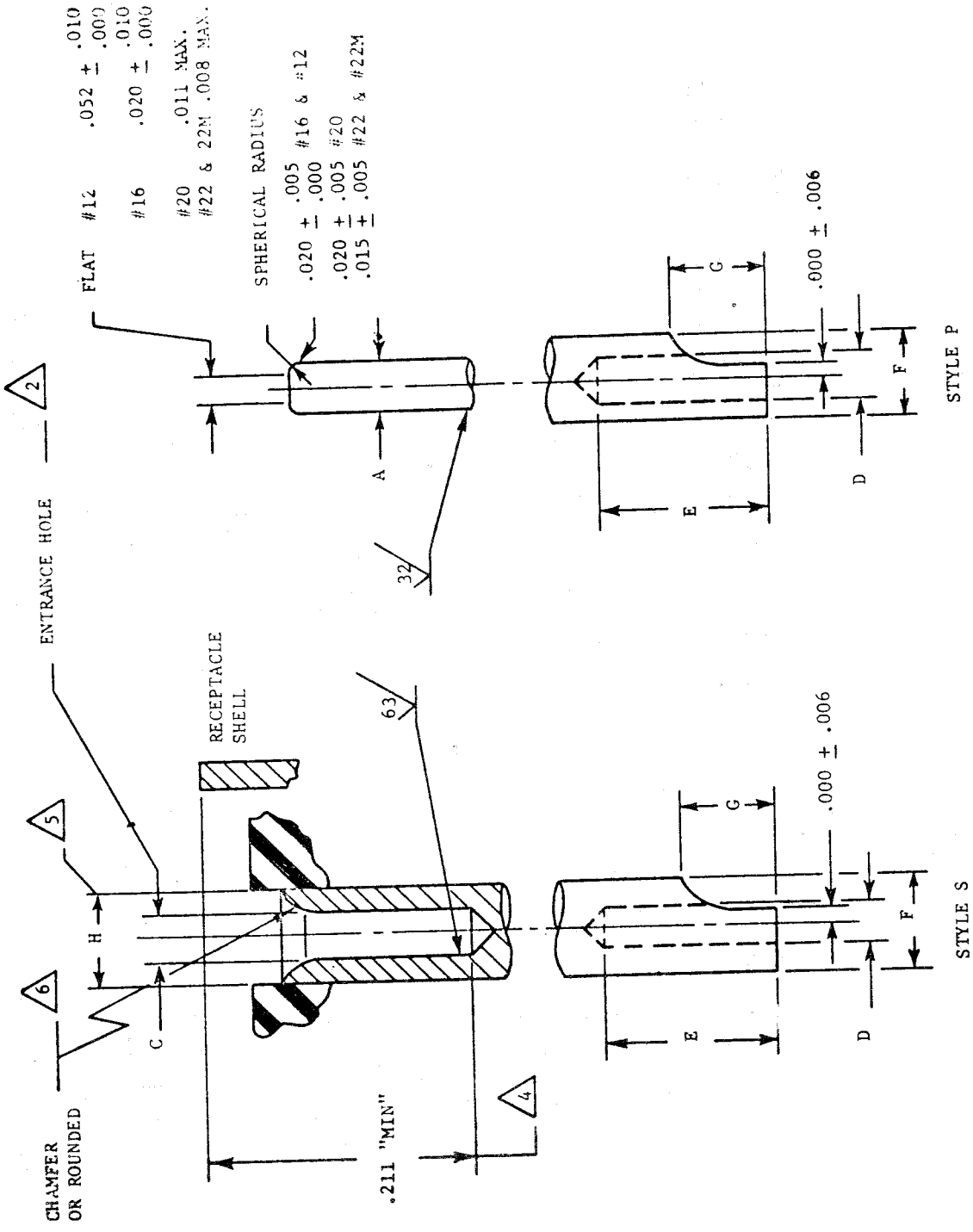


FIGURE 1 CLASS Y CONTACT CONFIGURATION & DIMENSIONS VARIETY A

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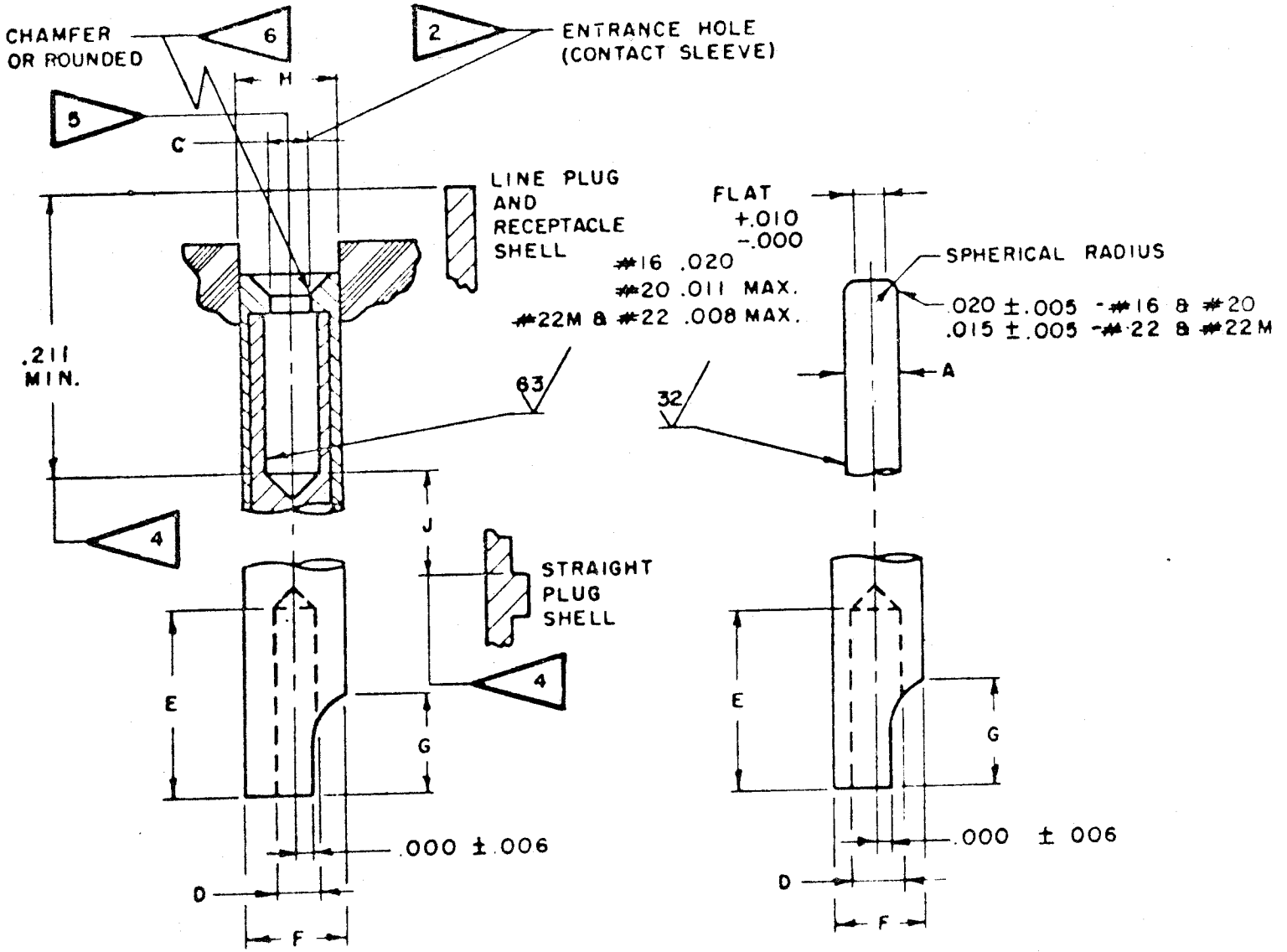


FIGURE 3 CONTACT CONFIGURATION & DIMENSIONS (Sheet 1) VARIETY B

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3.4.2.1 Variety A. For other than Class H, the entire insert and wire sealing or wire supporting member shall be essentially one integral part, designed to provide suitable sealing and support around the wires and to be nonremovable. The rigid dielectric shall be one molded piece. The design shall be such as to permit the removal and replacement of individual contacts by means of the MS27495 tools. The contact locking device shall be contained in the insert and shall so retain the contacts as to meet the contact retention requirements of this specification. Inserts shall be secured to prevent rotation. Vitreous material shall be used to insulate and seal contacts in Class H. assemblies. The insert member for Class H, Style S shall be of two-piece construction consisting of a reinforced thermosetting plastic socket support member bonded to the face of the vitreous insert and mechanically retained in the shell. All pin contact inserts shall have a resilient interface seal bonded to the front face in accordance with the applicable standards. Socket entry holes and pin "donut" rings shall conform to figure 2.

3.4.2.2 Variety B - Inserts (except for Class H, Style S), shall be molded in one piece and shall be free from voids or surface irregularities and shall have mold identification. Class H, Style S shall be two-piece construction with all interfaces bonded. For Classes T and P, the mold identification number shall appear on the rear face (solder cup side) of the insert.

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3.4.3 Sealing - Variety A

3.4.3.1 Wiring Sealing. Class E and T assemblies shall be designed to meet the environment requirements of this specification, using wire of outer diameter within the applicable range, as shown following:

Contact Size	Wire Size (AWG)	Finished Wire OD (Inch) Minimum	Maximum
22M	28,26,24	0.030	0.050
22	26,24,22	0.034	0.060
20	24,22,20	0.040	0.083
16	20,18,16	0.066	0.109
12	14,12	0.097	0.142

Connectors shall meet the requirements specified when:

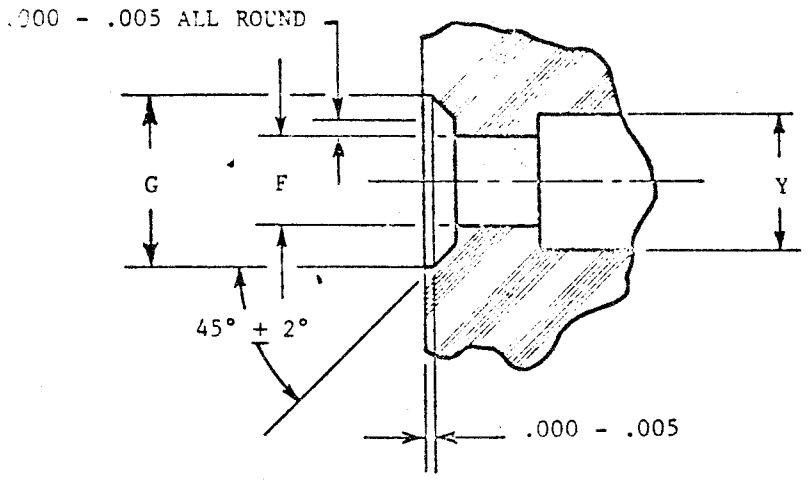
a. A full complement of wire of the applicable minimum or maximum insulation diameter is installed.

b. Any combination of wire diameters within the extremes of a, preceding are used.

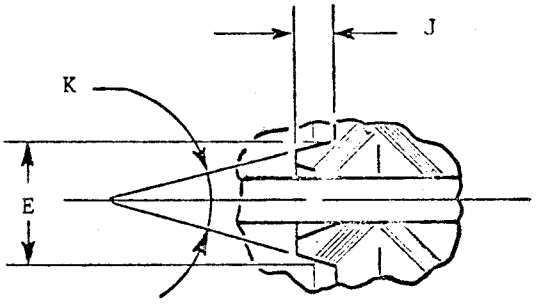
NOTE: For grommeted assemblies with followers, the total cumulative cross-sectional area of the wire bundle must not exceed the area resulting from a full complement of wires having the following diameter: 0.044, 0.052, 0.072, 0.095, and 0.126 for contact sizes 22M, 22, 20, 16 and 12, respectively.

3.4.3.2 Mating Seal. Plugs and receptacles with pin inserts shall have a resilient face with individual pin barriers (fig. 2). The pin barrier projections shall seal in their respective lead-in chamfers of the hard face socket insert. The resilient interfacial seal shall provide individual contact seals in the mated condition to ensure circuit isolation between each contact and contact to shell. The receptacle shall be provided with a peripheral seal.

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DETAIL A SOCKET ENTRY



DETAIL B PIN "BARRIER"

	DETAIL B			DETAIL A			
CONTACT SIZE	E DIA.	J	K	F DIA.	G DIA.	Y DIA.	
22M & 22	.077 / .069	.040 / .030	28° / 26°	.038 / .035	.067 / .063	.0665 / .0640	
20	.099 / .091	.040 / .030	28° / 26°	.052 / .049	.089 / .085	.0860 / .0830	
16	.122 / .114	.040 / .030	28° / 26°	.074 / .071	.112 / .108	.1210 / .1180	
12	.154 / .146	.040 / .030	28° / 26°	.106 / .103	.144 / .140	.1690 / .1660	

FIGURE 2 DETAIL B PIN BARRIER VARIETY A

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3.4.4 Shell. Shell shall be of one-piece construction and shall be designed to retain their inserts in one position, both axially and with respect to rotation, by mechanical means. Adhesives shall be used as a supplementary retention means for Classes E, P, and T, Variety A, and for Classes T and P, Variety B.

3.4.4.1 Grounding Spring. When applicable, springs shall have a minimum of six fingers per inch. Spring fingers shall be designed to make electrical contact with the mating shell without interfering with proper engagement. The grounding springs shall be permanently fixed to the shell periphery by riveting, welding, brazing, or soldering. For all assemblies except Variety A, Series II, the spring fingers shall engage the mating shell at a minimum distance of .090-inch prior to pin-socket engagement. For Variety A, Series II assemblies, the spring fingers shall engage the mating shell after the pins engage the sockets.

3.4.5 Screw threads. Screw threads shall be in accordance with MIL-S-7742. Out-of-roundness is not objectionable if the threads can be checked without forcing the thread gages.

3.4.6 Coupling. Connectors shall be coupled to counterpart connectors by means of a bayonet mechanism. The mechanism shall include a means of maintaining the mated connector in full engagement. The coupling ring shall be knurled or fluted to facilitate coupling. Coupling pins shall be riveted, welded, or brazed in place.

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3.4.6.1 Ease of Coupling. Counterpart connectors of any arrangement shall be capable of being fully coupled and uncoupled in a normal and accessible location without the use of tools.

3.4.6.2 Positive Detent. A detent shall be provided in the coupling mechanism so that an audible click is evident when proper coupling is accomplished. Provision shall be made for visual determination that proper coupling has been achieved.

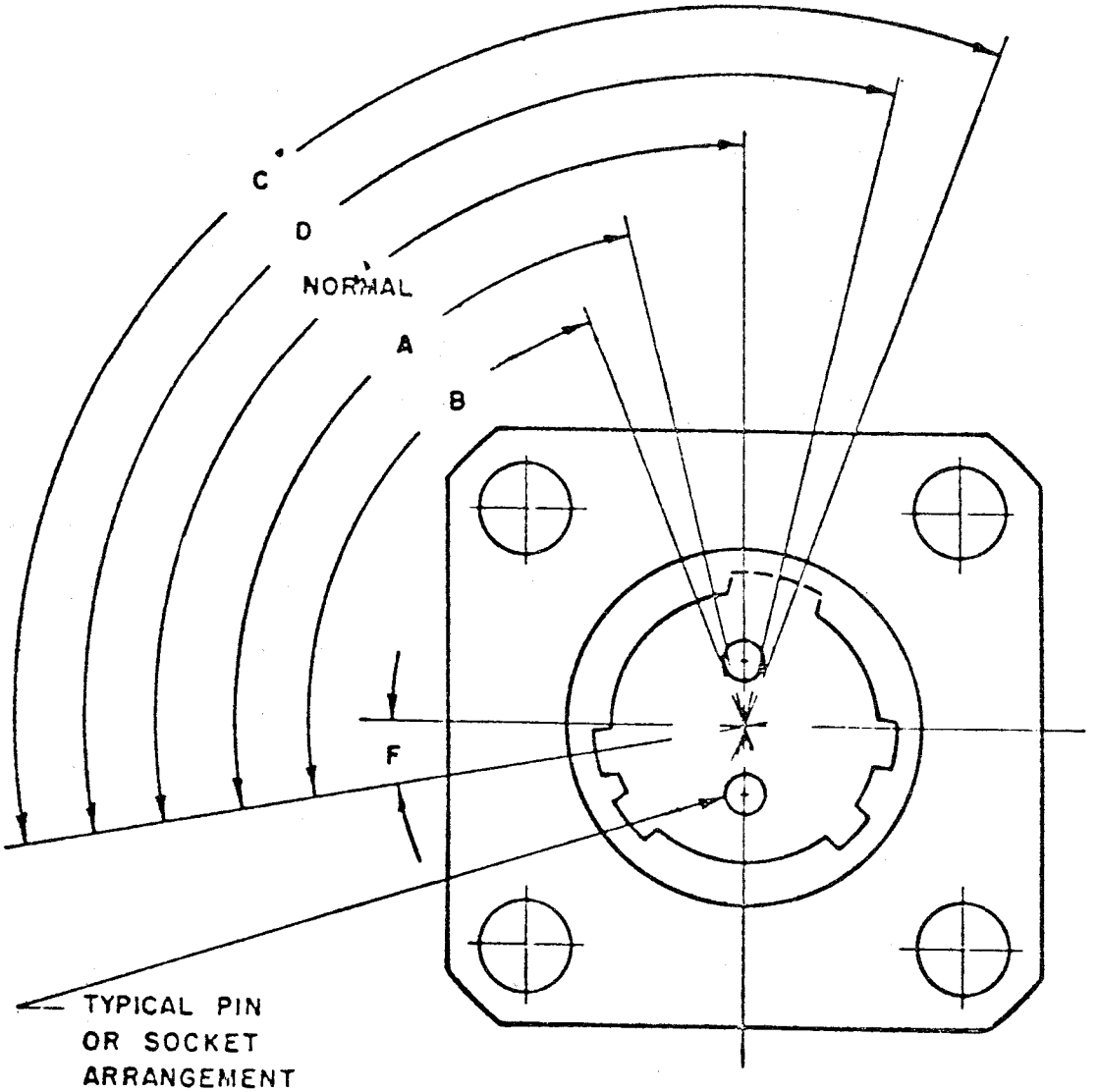
3.4.6.3 Polarization. Polarization of connectors shall be accomplished by means of five integral keys and suitable matching keyways on the counterpart. Polarization of mating connectors shall be accomplished before initial engagement of the coupling ring is possible. During axial engagement, pins shall not contact sockets or the insert face until polarization has been achieved.

3.4.6.4 Alternate (Keyway) Positions. Position of shell shall also be provided with the main key (or keyway) rotated from the normal as indicated in figure 4. The relationship between the four small keys (or keyway) and the insert arrangement centerline shall remain constant.

3.4.6.5 Lubrication

3.4.6.5.1 Variety A - Bayonet coupling ramps, spring washers, and threads shall be coated with a suitable lubricant. Rear accessory threads of Classes P and T shall not be lubricated.

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NOTE: FRONT FACE OF RECEPTACLE SHOWN. PLUG POLARIZATIONS ARE MIRROR IMAGE.

FIGURE 4 MAIN KEYWAY POLARIZATION (SHEET 1)

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SERIES I ASSEMBLIES

Varieties A and B

Shell	F	Normal	Suffix	Alternates Letter	Designation	
	REF		A	B	C	D
9	5°	95°	77°	--	--	113°
11	5°	95°	81°	67°	123°	109°
13	5°	95°	75°	63°	127°	115°
15	5°	95°	74°	61°	129°	116°
17	5°	95°	77°	65°	125°	113°
19	5°	95°	77°	65°	125°	113°
21	5°	95°	77°	65°	125°	113°
23	5°	95°	80°	69°	121°	110°
25	5°	95°	80°	69°	121°	110°

SERIES II ASSEMBLIES

Varieties A and B

Shell	F	Normal	Suffix	Alternates Letter	Designation	
	REF		A	B	C	D
8	10°	100°	82°	--	--	118°
10	10°	100°	86°	72°	128°	114°
12	10°	100°	80°	68°	132°	120°
14	10°	100°	79°	66°	134°	121°
16	10°	100°	82°	70°	130°	118°
18	10°	100°	82°	70°	130°	118°
20	10°	100°	82°	70°	130°	118°
22	10°	100°	85°	74°	126°	115°
24	10°	100°	85°	74°	126°	115°

FIGURE 4 MAIN KEYWAY POLARIZATION (SHEET 2)

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3.4.6.5.2 Variety B - Bayonet coupling ramps, spring washer housing, and all threads except those intended for potting compound anchorage shall be coated with a lubricant conforming to MIL-G-3545.

3.4.7 Cavity Fill

3.4.7.1 Variety A - Classes E, P, and T. The cavity at the rear of the connectors between the insert and the shell shall be filled with RTV silicone compound or equivalent.

3.4.7.2 Variety B - Class T, Series I and II; Class P, Series I. - The cavity at the rear of the connectors between the insert and the shell shall be filled with a RTV silicone compound or equivalent. The compound shall form a continuous bond, shall be visible, and shall not impair the legibility of the contact lettering.

3.4.8 Plating

3.4.8.1 Variety A (Classes E, P, and T). Contacts for Series I and IIA shall be silver-plated a minimum of 0.00015-inch thick to QQ-S-365 followed by gold plating to a minimum of 0.00005-inch thick to MIL-G-45204. Series IIB contacts shall be silver-plated to a minimum of 0.0001-inch thick to QQ-S-365 followed by gold plating to a minimum of 0.0001-inch thick to MIL-G-45204.

3.4.8.2 Variety A Contacts (Class H). When used in assemblies with carbon steel shells, contacts shall be tin-plated a minimum 0.0001-inch thick over 0.0002-inch minimum copper plate. The tin plate shall be reflowed to retain solderability. When used in assemblies with corrosion-resistant steel shells, contacts shall be gold-plated to a minimum thickness of 0.0001-inch to MIL-G-45204 over a minimum thickness of 0.0001-inch of silver plate to QQ-S-365.

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3.4.8.3 Variety B Contacts (Classes T and P). Contacts for Series I shall be silver-plated a minimum of 0.0001-inch thick to QQ-S-365, followed by gold plating to a minimum of 0.0001-inch thick to MIL-G-45204. Series II contacts shall be silver-plated to a minimum of 0.00015-inch thick to QQ-S-365 followed by gold plating to a minimum of 0.00005-inch thick to MIL-G-45204.

3.4.8.4 Variety B Contacts (Class H). When used in assemblies with carbon steel shells, contacts shall be tin-plated in accordance with MIL-T-10727 a minimum of 0.0001-inch thick over 0.0002-inch minimum copper plate. The tin plate shall be reflowed to retain solderability. When used in assemblies with corrosion resistant steel shells, contacts shall be gold-plated to a minimum thickness of 0.0001-inch to MIL-G-45204 over a minimum thickness of 0.0001-inch of silver plate to QQ-S-365.

3.4.8.5 Variety A Shells and Accessory Hardware. Aluminum alloy parts for Series I and IIA connectors shall be nickel-plated a minimum of 0.0002-inch followed by cadmium plate 0.0001-inch minimum in accordance with QQ-P-416, Type II. A preliminary plating of other metal is permissible. The final finish shall be electrically conductive and shall be bright or light iridescent (Finish A) or olive-drab (Finish B). Aluminum alloy parts for Series IIB connectors shall have a hard, anodic, nonconductive finish, (Finish C) to MIL-F-14072, Finish E516. Carbon steel shells shall be tin-plated (Finish D) in accordance with MIL-T-10727 and the tin shall be reflowed to promote solderability.

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3.4.8.6 Variety B Shells, Coupling Ring, Hex Jam Nut, Potting Ring and Accessory Hardware. Aluminum alloy parts shall be nickel-plated a minimum of 0.0002-inch followed by cadmium flash 0.0001-inch minimum in accordance with QQ-P-416, Type II. A preliminary plating of other metal is permissible. The final finish shall be electrically conductive and shall be bright or light iridescent (finish D).

3.4.8.7 Variety A Gounding Spring. Grounding springs shall be plated to prevent corrosion.

3.4.8.8 Variety B Grounding Spring. Grounding springs for Series I assemblies shall be plated to prevent corrosion.

3.5 Interchangeability. All connectors having the same part number shall be completely interchangeable with each other with respect to installation and performance.

3.5.1 Physical Requirements. In addition, connectors shall be capable of withstanding tests simulating rough usage incidental to handling, transportation, and service use.

3.6 Performance.

3.6.1 Environmental Requirements. Connectors shall be capable of satisfactory performance during, or after, as applicable, subjection to the following environmental conditions:

a. Temperature

(1) Variety A. Temperatures from -55°C (-67°F) to 150°C (302°F) for Series I and IIA connectors. Temperatures from -65°C (-85°F) to 200°C (392°F) for Series IIB connectors with the exception of solder-mount receptacles which have an upper limit og 150°C.

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(2) Variety B. Temperature from -55°C (-67°F) to 177°C (351°F) for Series I to 150°C (302°F) for Series II including frequent cycling between these extremes under conditions of thermal shock.

b. Vibration from 10 to 2,000 Hz sinusoidal at a level of 40g or total excursion of 0.06-inch amplitude, whichever is the least severe. Vibration from 2,000 to 3,000 Hz sinusoidal at a level of 20.

c. Altitudes from sea level to 110,000 feet (0.21 in Hg).

d. Shock as represented by a 0.5-millisecond base half-sine wave peaking at 1,500 g.

e. Humidity conditions up to 100 percent RH including condensation.

f. Exposure to salt-laden atmosphere typifying marine or coastal conditions.

g. Thermal Shock. Mated connectors shall be capable of withstanding thermal shock from -55°C (-67°F) to 150°C (302°F) minimum.

h. Solvent Immersion. Unmated connectors shall be immersed in jet fuel, lubricating oil, and hydraulic fluid, as specified.

i. Acceleration. Mated connectors shall be capable of withstanding an acceleration from 0 to 200 g in one minute along each of three mutually perpendicular axes.

j. Random vibration from 100 Hz to 3,000 Hz with a power spectral density of 1 g^2/Hz, attenuated at not less than 12 decibels (db) per octave below 50 Hz and above 3,000 Hz, with an over-all rms G of 54.3 + 15 percent, -5 percent.

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k. Acoustical Noise. Mated connectors shall be capable of withstanding an over-all sound pressure level of 155 db.

3.6.2 Grounding Spring Force (Variety A and Variety B, Series I only). Grounded plugs and receptacle shells, the latter provided without bayonet pins, shall be engaged and separated to their fullest extent. On the initial mating, the forces necessary to engage and separate shall satisfy the Ground Spring Force Test for this specification.

3.6.3 Shell Resistance (Variety A and Variety B, Series I only). The electrical resistance shall be expressed as potential drop. The potential drop measured across a mated assembly from the rear end of the slug shell to a point on the receptacle flange shall not exceed 50 millivolts when a steady-state current of 10 amperes dc is applied.

3.6.4 Maintenance Aging (Except Variety A, Class H and Variety B, Class H). Connectors shall be capable of meeting the performance requirement of this specification after having been subjected to several cycles of connector mating and unmating and insertion and withdrawal of contacts using the appropriate tools. To simulate accelerated service usage, the Maintenance Aging Test of this specification shall be applied.

3.6.5 Insert Retention. Assemblies shall retain their inserts in their proper location in the shell when subjected to a pressure of 100 psig.

3.6.6 Contact Retention. Contacts in connectors shall withstand an axial force when tested as specified herein.

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3.6.7 Coupling Pin Strength. Bayonet coupling pins shall withstand a load of 50 pounds applied to the attached end along the longitudinal axis of the pins without displacement or preceptible loosening of the coupling pin resulting from this test.

3.6.8 Engaging and Separating Forces. Engaging and separating forces of socket contacts shall be within the values of the contact engaging and separating force test of this specification.

3.6.9 Test-Probe Damage. Socket contacts shall withstand the bending moments specified in the Test Probe Damage Test of this specification when such moments are applied. Contact resistance shall be less than 3 milliohms before and after testing.

3.6.10 Dielectric Withstanding Voltage. Connector shall show no evidence of flashover or breakdown when the applicable test voltage is applied for two seconds minimum.

3.6.11 Coupling Torque. The torque required to mate and unmate connectors shall meet the requirements for the appropriate shell size specified in the Coupling Torque Test of this specification.

3.6.12 Durability. Connectors shall be capable of being mated and unmated 250 times at a rate not exceeding 200 cycles per hour.

3.6.12.1 Contact Durability. There shall be no discoloration of pin contacts within the first 200 mating/unmating cycles when tested as specified herein.

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3.6.13 Contact Resistance. The resistance of two mated pairs of contacts shall be such that the potential drop at 1 ampere shall not exhibit a change greater than 4 millivolts for two pairs of mated contacts when tested in accordance with the contact resistance tests of this specification.

3.6.14 Insulation Resistance

3.6.14.1 Variety A

3.6.14.1.1 Insulation Resistance at Room Temperature. The insulation resistance between any pair of contacts and between any contact and the shell shall not be less than 100,000 megohms for Classes E, P, and T and 50,000 megohms for Class H.

3.6.14.1.2 Insulation Resistance at Elevated Temperature. For Classes E, P, and T, the insulation resistance shall not be less than 1,000 megohms for Series I and IIA, and 200 megohms for Series IIB, when tested in accordance with the Insulation Resistance at Elevated Temperature Tests of this specification. Similarly, for Class H connectors, the insulation resistance shall not be less than 100 megohms for Series I and IIA, and 10 megohms for Series IIB.

3.6.14.2 Variety B

3.6.14.2.1 Insulation Resistance. The insulation resistance between any contact and all other contacts in common with the shell shall not be less than 100,000 megohms for Classes T and P, and 50,000 megohms for Class H.

3.6.15 Air Leakage

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3.6.15.1 Variety A

3.6.15.1.1 Class H. Receptacles, when subjected to a pressure differential of one atmosphere across the connector, shall not exhibit an air leakage rate of more than 0.1-micron cubic foot per hour (1×10^{-6} cc/sec).

3.6.15.2 Variety B

3.6.15.2.1 Class T (all) and Class P. Receptacles shall show no visible evidence of leakage when subjected to a pressure differential of one atmosphere across the connector.

3.6.15.2.2 Class H. Receptacles, when subjected to a pressure differential of one atmosphere across the connector, shall exhibit an air leakage rate of not more than 0.1-micron cubic foot per hour (1×10^{-6} cc/sec).

3.6.16 Altitude Immersion (Variety A only). Mated connectors shall retain an insulation resistance greater than 2,000 megohms and shall withstand a high potential test using the applicable sea level values.

3.7 Reliability. The supplier's reliability program for assembled connectors and assembly procedures shall meet the requirements of MIL-STD-790.

3.7.1 Reliability Level. Successful completion of the Qualification Tests as herein described shall constitute the lowest level of reliability to which the subject connectors will be procured. Higher levels of reliability may be attained by Life Testing.

3.7.1.1 Reliability Level Indicator. Each connector supplied in accordance with this specification shall be marked with a one-digit code. Marking shall be in accordance with 3.8. The one-digit code will indicate the reliability level that has been demonstrated by the supplier in conformance with the requirements of 4.8.8.

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3.7.1.2 Higher Reliability Levels. Higher reliability levels may be attained only by demonstration that the connector meets the requirements of the Reliability Life Tests. Life testing may be accumulative.

3.8 Identification of Product. Connectors shall be metal or ink-stamped with the manufacturer's name or trademark and the following information in accordance with MIL-STD-130 where space permits. Metal stamping may be accomplished before or after plating. The following examples are illustrative:

a. Identification

____	E	A	13	____	8	P	A	2
Slash Number	Variety	Finish	Shell Size		Insert Angle		Polarization	Style Kind

b. Lot Number - 000010

3.8.1 Insert Identification. Where space permits, the insert number, e.g., 17-26, shall appear on the front face of the insert in a contrasting color.

3.8.1.1 Contact Location Identification. Contact locations shall be identified as indicated on the applicable specification sheet. All positions shall be identified on the front and rear faces of the insert except where space limitations make this impracticable. Location of contact identifying characters shall be in close proximity to the holes, but need not be placed exactly where indicated on the specification sheet.

3.8.2 Contact Identification. Crimp contacts shall be identified with the manufacturer's trademark or other suitable identification.

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3.9 Workmanship. The connector shall be fabricated in a manner such that the criteria for appearance, fit, and adherence to specified tolerances are observed. Particular attention shall be given to neatness and thoroughness of marking of parts, plating, welding, soldering, riveting, staking, bonding, and freedom of parts from burrs and sharp edges which might mar adjacent equipment or cuase injury to operating personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification. Inspection and testing of connectors to this specification shall consist of:

a. Qualification Tests

b. Acceptance Tests

c. Reliability Life Tests (levels 2, 3, 4, and 5)

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4.2.1 Qualification. The object of qualification testing is to confirm that the supplier has the capability to manufacture products which meet the requirements of this specification. Qualification approval may be granted upon successful completion of the inspections and tests of 4.2.1.3, conducted on the samples of 4.2.1.1; such approval to include classes, finishes, shell configurations, and insert arrangements other than those actually tested provided that the capability to manufacture these parts is demonstrated by providing exhibit samples.

4.2.1.1 Qualification Test Samples. The following test samples for each variety and series, manufactured by normal production methods, shall be provided.

4.2.1.1.1 Variety A

a. Two mating pairs, Class E straight plug and wall mounting receptacle, of each of the nine shell sizes. Three shell sizes shall be Service M, three shall be Service I, and three shall be Service II. Each arrangement in a given shell size shall be the densest contact arrangement for that particular service rating. One sample of each shell size shall have pin contacts in the receptacle and socket contacts in the plug. The other sample shall have socket contacts in the receptacle and pin contacts in the plug.

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b. Two mating pairs, Class Y jam-nut receptacle and Class E straight plug, of each of the nine shell sizes. Three shell sizes shall be Service M, three shall be Service I, and three shall be Service II. Each arrangement in a given shell size shall be the densest contact arrangement for that particular service rating. One sample of each shell size shall have pin contacts in the receptacle and socket contacts in the plug. The other sample shall have socket contacts in the receptacle and pin contacts in the plug. Suppliers seeking qualification of one style only in Class Y shall provide both jam-nut receptacles in that style with mating Class E counterparts.

c. Fifty crimp-type contacts of each size and type and sufficient additional plugs and receptacles to provide fifty Class Y pin and socket contacts of each size.

d. Two Series I and II wall mount receptacles in each shell size, less coupling pins and inserts.

e. Two Series II straight grounded plug shells in each shell size.

f. Two mating connectors of each of the following:

(1) Straight plug and wall mount receptacle of each Series in Class E.

(2) Straight plug, Class E, and jam-nut receptacle Class Y in each Series.

The connectors shall be in two different shell sizes, but all three Service Ratings shall be represented.

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4.2.1.1.2 Variety B

a. Two mating pairs, Class T straight plug and wall mounting receptacle of each size. Three shell sizes shall be Service M, three shall be Service I, and three shall be Service II. Each arrangement in a given shell size shall be the densest contact arrangement for that particular service rating. One sample of each shell size shall have pin contacts in the receptacle and socket contacts in the plug. The other sample shall have socket contacts in the receptacle and pin contacts in the plug.

b. Two mating pairs, Class H jam-nut receptacle with Class T or P straight plug of each shell size. Three shell sizes shall be Service M, three shall be Service I, and three shall be Service II. Each arrangement in a given shell size shall be the densest contact arrangement for that particular service rating. One sample of each shell size shall have pin contacts in the receptacle and socket contacts in the plug. The other sample shall have socket contacts in the receptacle and pin contacts in the plug. Suppliers seeking qualification of one style only in Class H shall provide both jam-nut receptacles in that style with mating Class T or P counterparts.

c. Additional plugs and receptacles to provide a minimum of fifty each pins and sockets of each size in Classes T and H.

d. Two Series I wall mount receptacles in each shell size, less coupling pins and inserts.

e. Two Series II straight grounded plug shells in each shell size.

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f. Two mating connectors of each of the following:

(1) Straight plug and wall mount receptacle of each Series in Class T.

(2) Straight plug, class T and jam-nut receptacle Class H in each Series.

The connectors shall be in two different shell sizes, but all three Service Ratings shall be represented.

4.2.1.2 Preparation of Samples. Wiring of crimp-type contacts shall be accomplished using tools per MS3198-1 and/or MS3191-4.

a. Samples, provided under 4.2.1.1.1a and 4.2.1.1.2 shall be wired with wire having a smooth extruded outer jacket of waterproff construction. Half the samples shall have wires as listed under column A in Table I; the other half shall have wires as listed under column B of the same table

Table I Test Wire Sizes

Contact Size	Col. A Max. Dia.	Col. B Min. Dia.
22M	MS21985-24	MS21985-28 MIL-W-16878/4-28
22	MS21985-22	MS21985-26 MIL-W-16878/4-26
20	MS18113-20 MIL-W-16878/5-20	MS21985-24 MIL-W-16878/4-24
16	MIL-W-16878/5-16	MS18113-20
12	MIL-W-16878/5-12	MS18113-14

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b. Samples provided under 4.2.1.1.1b and 4.2.1.1.2 shall be wired with wire conforming to MIL-W-22759 or MIL-W-16878/4 of nominal gage at any convenient stage in the sequence prior to vibration.

c. Samples of individual contacts provided under 4.2.1.1.1.c shall be crimped to silver-plated wire conforming to MIL-W-22759 or MIL-W-16878/4 or /5 so as to have 15 mating contacts of each contact/wire combination as follows:

Contact Size	22M			22			20			16		
Wire AWG	24	26	28	22	24	26	20	22	24	16	18	20

In addition, five mating contacts of each size shall be crimped to the smallest applicable wire gage.

d. Samples of solder contacts in Variety B, Classes T and H, and Variety A, Class H receptacles provided under 4.2.1.1.2 and 4.2.1.1.1.c shall be terminated to wires of nominal gage conforming to MIL-W-22759 or MIL-W-16878/4.

4.2.1.3 Qualification Tests

a. Samples provided under 4.2.1.1.1.a and 4.2.1.1.2a shall be divided into two numerically equal groups. Each group shall contain roughly the same number of samples with the larger diameter wire as with the smaller diameter wire. One group shall be subjected to the tests of Group I, Table II in the sequence indicated. The other group shall be subjected to the tests of Group II, Table II in the sequence indicated.

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b. Samples provided under 4.2.1.1.1b and 4.2.1.1.2b shall be divided into two similar groups. One group shall be subjected to the tests of Group I, Table II in the sequence indicated. The other group shall be subjected to the tests of Group III, Table II in the sequence indicated.

c. Samples provided under 4.2.1.1.1c and 4.2.1.1.2c shall be disposed as follows:

(1) Crimp-type individual contacts shall be divided into four subgroups. Subgroups I through III shall contain five mating contacts of each contact/wire combination. Subgroup IV shall consist of five mating contacts of each size, terminated to the smallest applicable gage wire.

(2) Solder contacts in Class Y, Variety A and Classes T and W, Variety B connectors shall be subjected to the tests of Group IV, Table II in the sequence indicated.

d. Samples provided under 4.2.1.1d shall serve as counterparts for the tests of 4.6.1 and 4.6.2.

e. Samples provided under 4.2.1.1e shall be subjected to the tests of 4.6.1 and 4.6.2, where applicable, to Series II connectors.

f. Samples provided under 4.2.1.1f shall be subjected to the tests of Group VI, Table II.

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TABLE II QUALIFICATION TESTS

	Test Paragraph
Group I (Classes E, P, T, and Y, Variety A; Classes T, P, and H, Variety B)	
Examination of Product	4.5
Grounding Spring Force (Grounded Plugs Only)	4.6.1
Shell Resistance (Grounded Plugs Only)	4.6.2
Maintenance Aging (Classes E and T only), Variety A and Class T, Variety B	4.6.3
Thermal Shock (Class Y only), Variety A; Class H, Variety B	4.6.19
Temperature Cycling (Classes E, P, and T), Variety A; Classes T and P, Variety B	4.6.8
Air Leakage (Class Y only), Variety A; Classes T and H, Variety B	4.6.21
Coupling Torque	4.6.12
Acceleration	4.6.25
Random Vibration	4.6.26
Acoustical Noise	4.6.14
Altitude Immersion (Class RE only), Variety A only	4.6.22
Insert Retention (Except Class Y), Variety A; Class H, Variety B	4.6.4
Corrosion	4.6.20
Coupling Torque	4.6.12
Contact Resistance	4.6.14
Solvent Immersion (Lube Oil only)	4.6.23
Coupling Torque	4.6.12
Coupling Pin Strength (Style P only)	4.6.6
Group II (Classes E, P, and T, Variety A; Classes T and P, Variety B)	
Examination of Product	4.5
Maintenance Aging (Classes E and T only, Variety A; Class T, Variety B)	4.6.3
Contact Retention	4.6.5
Insulation Resistance at Room Temperature	4.6.15.1 & 4.6.15.2
Temperature Cycling	4.6.8
Coupling Torque	4.6.12
Insulation Resistance at Elevated Temperature	4.6.15.1.2
Dielectric Withstanding Voltage	4.6.11
Durability	4.6.11
Shell Resistance (Grounded Plugs only)	4.6.13
Sinusoidal Vibration	4.6.2
Shock	4.6.16
Moisture Resistance	4.6.17
Solvent Immersion (JP-4 only)	4.6.18
Coupling Torque	4.6.23
	4.6.12

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TABLE II CONTINUED

	Test Paragraph
Group III (Class Y Receptacles	
Examination of Product	4.5
Thermal Shock	4.6.19
Air Leakage	4.6.21
Durability	4.6.13
Coupling Torque	4.6.12
Sinusoidal Vibration	4.6.16
Shock	4.6.17
Dielectric Withstanding Voltage, Sea Level only	4.6.11
Insulation Resistance at Room Temperature	4.6.15.1 & 4.6.15.2
Moisture Resistance	4.6.18
Contact Resistance	4.6.14
Solvent Immersion (JP-4 only)	4.6.23
Coupling Torque	4.6.12
Group IV Variety A, Class H; Variety B, Class H (Contacts Mated with Crimp Counterparts)	
Contact Resistance	4.6.14
Engaging and Separating Forces	4.6.9
Test-probe Damage	4.6.10
Engaging and Separating Foces	4.6.9
Contact Resistance	4.6.14

Group V (Crimp Contacts)

	Subgroups I	II	III	IV	
Examination of Contact	X	X	X	X	4.7.1
Crimp Deformation	X		X		4.7.2
*Temperature Durability	X				4.7.3
Corrosion				X	4.6.20
Contact Resistance	X	X		X	4.6.14
Test-probe Damage		X			4.6.10
Engaging and Separating Forces		X	X		4.6.9
Contact Resistance		X			4.6.14
Durability Conditioning		X			4.7.4
Corrosion		X			4.6.20
Contact Resistance		X			4.6.14
Tensile Strength			X		4.7.5
Examination of Contact	X	X	X	X	4.7.1

*Qualification only

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TABLE II CONTINUED

	Test Paragraph
Group VI (Classes E, P, T and Y, Variety A; Classes T, P, and H Variety B)	
Examination of Product	4.5
Temperature Durability	4.6.24
Coupling Torque	4.6.12
Dielectric Withstanding Voltage	4.6.11
Insulation Resistance, Room Temperature	4.6.15.1 & 4.6.15.2
Insulation Resistance at Elevated Temperature	4.6.15.1.2
Insert Retention	4.6.4
Contact Retention	4.6.5
Air Leakage (Class Y, Variety A; Class H, Variety B)	4.6.21
Solvent Immersion (Hydraulic Fluid only)	4.6.23

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4.2.2 Acceptance Tests. The object of acceptance testing is to assure that the standards of quality and performance required for products procured to this specification and demonstrated by qualification testing are being maintained in current production. Acceptance testing shall consist of the following:

a. Individual tests

b. Sampling Tests

c. Degrading tests

4.2.2.1 Individual Tests. Each connector shall be subjected to the individual tests shown in Table III, Group I.

4.2.2.2 Sampling Tests.

4.2.2.2.1 Lot Definition. The production lot definition of MIL-STD-790 is further defined as follows:

a. A production lot consists of connectors conforming to the same number, manufactured from the specified raw materials, processed under the same specification and procedures, produced by the same type of equipment, and submitted for inspection at one time. Each production lot of assembled connectors shall be a group identified by a common manufacturing record through all significant assembly operations.

b. Traceability of connectors to specific physical/chemical test reports of incoming raw material is not required.

c. Common manufacturing records and traceability shall begin with the start of connector assembly.

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4.2.2.2.2 Tests. Connectors shall be selected from the production lot in accordance with MIL-STD-105D. The sample shall be subjected to the tests of Table III, Group II. Major and minor defects shall be as defined in MIL-STD-105.

4.2.2.3 Degrading Tests. The samples shall be randomly selected from the lot and subjected to the degrading tests of Group III in the sequence listed. Samples shall be tagged or otherwise identified as degraded test samples and shipped. Samples shall be selected as shown following:

Lot Size	Sample Size
1-50	1
51-100	2
101-200	4
201-400	8
Over 400	2% of lot size

4.2.2.3 Additional sample pieces shall be built with the lot in accordance with the above schedule.

4.2.2.4 Rejection and Retest. When a sample item fails to meet the specified tests, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.2.2.4.1 Individual Tests May Continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure; but final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of this specification.

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TABLE III

ACCEPTANCE TESTS

Contacts shipped with connectors other than Class Y, Variety A, shall be from lots that have met the requirements of 4.2.3.

	Test Para.	AQL	NOTES
Group 1 - Individual Tests			
Examination of product - 100% visual examination per 4.5 and 100% inspection of critical characteristics as defined in			1/
Dielectric Withstanding voltage (sea level only) (Except Class Y, Style P)	4.6.11	100%	
Insulation Resistance (room temperature)	4.6.15.1.1 and 4.6.15.2	100%	4/
Group 2 - Sampling Tests	**Test Para.**	**AQL**	
Examination of Product	4.5		
Major		0.25	1/
Minor		1.0	1/
Tests			
Air Leakage (Class Y Variety A only)	4.6.21	0.25	1/
Engaging and Separating Forces (Class T, Variety B)	4.6.9	0.25	1/
Contact Resistance	4.6.14	1.0	3/
Grounding Spring Force	4.6.1		
Coupling Pin Strength	4.6.6	1.0	1/

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TABLE III Continued

Group 3 - Degrading Tests	Test Para.	NOTES
Axial position of pin contacts, Point of spring engagement, socket contact	Slash Sheets	
Temperature cycling (Classes E, P, and T, Variety A; Classes T and P, Variety B)	4.6.8	5/
Air Leakage (Class Y, Variety A; Class H, Variety B)		
Coupling Torque (Plugs only, except line plugs	4.6.12	
Test-probe damage (Class Y, Variety A; Class H, Variety B)	4.6.10	
Engaging and Separating Forces (Class Y, Variety A; Class H, Variety B, Style S only)	4.6.9	
Insert Retention (Classes E, P, and T, Variety A; Classes T and P, Variety B)	4.6.4	
Contact Retention (Classes E, P, and T, Variety A; Classes T and P, Variety B)	4.6.5	

NOTES

1/ The supplier may use in-process controls for this requirement.

2/ Test 5 pieces - no failures permitted.

3/ Select sample connectors in accordance with the AQL levels shown. Test 3 contacts in each sample connector.

4/ Test between two adjacent contacts and between two peripheral contacts and the shell.

5/ Sample connectors need not be mated for this test.

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4.2.2.5 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.2.3 Variety A

Acceptance Tests-Crimp-Type Contacts. Acceptance testing of crimp-type contacts shall consist of the following:

a. Sampling Tests

b. Periodic Tests

4.2.3.1 Sampling Tests

4.2.3.1.1 Lot Definition. A lot shall consist of a quantity of contacts, conforming to the same part number, manufacturered by the same processes and by the same type of equipment, and presented for acceptance at the same time.

4.2.3.1.2 Tests

a. Contacts shall be selected from each lot in accordance with MIL-STD-105, or MIL-STD-414 and subjected to Examination of Contact 4.7.1 - AQL 1.0%.

b. Samples shall be selected from each lot of socket contact in accordance with MIL-STD-105, or MIL-STD-414 and subjected to the following tests in the order listed:

(1) Engaging and Separating Forces - 4.6.9 - AQL 0.25%

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(2) Test-probe Damage - 4.6.10 - AQL 0.25%

(3) Engaging and Separating Forces - 4.6.9 - AQL 0.25%

In-process controls, unrelated to lot sizes, may be used provided the equivalent or tighter AQL is maintained.

4.2.3.2 Periodic Tests. Every three months, fifty mating crimp-type contacts of each size, representative of current production, shall be subjected to the applicable tests of Group 5, Table II. Preparation and disposition of samples shall be in accordance with QPL Instructions.

4.3 Test Equipment. Functional gages and special test equipment may be used for the examinations and tests specified herein.

4.3.1 Calibration and Accuracy. Except for humidity parameters, vibration amplitudes, shock/vibration accelerations, and discontinuity duration, test conditions shall be accomplished with equipment having accuracies within ±5 percent of their range. Calibration of test equipment shall be to Government standards. Calibration records shall be kept and be made available to the qualifying activity for review and approval on demand. All test chambers shall have a minimum volume of twice the volume of test items.

4.4 Test Conditions

4.4.1 General. Unless otherwise specified, all tests shall be performed at room atmospheric conditions. All items to be tested shall be at room temperature at the start of each test.

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4.5 Examination of Product. Each connector shall be visually examined for completeness, workmanship, and identification requirements. Attention shall be given to those assemblies that require a gasket to determine the conditions of that gasket. Gaskets missing, twisted, buckled, kinked, or damaged in any way shall be cause for rejection. Each Class E, P, or T, Variety A, and each Class T connector, Series I and II, and Class P connector, Series I, Variety B, connector shall be visually inspected for the presence of compound between wire termination end of insert and shell.

4.5.1 Detailed Inspection Detailed inspection shall be as follows:

a. Critical. The following dimensions shown on the pertinent specification sheets shall be verified 100 percent.

D. Diameter (max. on straight plugs, min. on line plug and receptacles

Key or keyway positions

Contact pattern relative to datum features. (Class H only)

b. Major

Identification

Loose coupling pins

Axial location of contacts (Class H, Style P only)

Point of spring engagement (Class H, Style S only)

c. Minor

(1) Insert-voids, chips, or open lines greater than one-quarter of contact-to-contact or contact-to-edge spacing.

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4.6 Test Methods (Connectors)

4.6.1 Grounding Spring Force. Plugs shall be completely mated and unmated with counterpart receptacles less coupling pins and inserts. On the initial mating and unmating, the forces necessary to engage and separate shall be within the following values.

Shell Size	Axial Force in Pounds		
	Max.	Min Var A	Min. Var. B
8 through 11	15	1/2	2
12 through 15	20	1/2	2
26 through 25	30	1/2	2

Failure to meet this requirement shall be cause for rejection.

4.6.2 Shell Resistance Straight plugs or plug shells Series I and IIA shall be mated with counterpart receptacles less coupling pins and inserts. Series IIA grounded connectors shall be mated and tested. Series I receptacles shall have a layer of dielectric material on the O.D. of the shell to prevent the coupling ring from making electrical contact with the receptacle shell, and an insulating washer of suitable I.D. and O.D 0.195±0.005 inch thick shall be installed on the plug shell skirt to prevent full engagement of the plug with the receptacle. A test current of 1+0.1 amp DC shall be caused through the shells of the mated assembly, using the voltmeter-ammeter method. The voltage drop between the rear end of the plug shell and a point on the front face of the counterpart flange shall be determined. A measured potential drop of more than 5 millivolts shall be cause for rejection.

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4.6.3 Maintenance Aging. Before the connector halves are mated/unmated, the contacts shall be mated and unmated 10 times. After the last mating/unmating cycle, the contacts will be removed and reinserted nine additional times. With all contacts in place, an axial load of 20 $^{+1}_{-0}$ lb shall be applied to both the pin and socket contacts in unmated connectors from the mating end. An initial onehalf pound preload shall be applied to take up the slack of the contact. The axial load shall be increased at 1 lb/sec until the specified 20-pound load is applied, or until the contact moves axially a distance greater than 0.012 inch. Axial contact movement greater than 0.012 inch or evidence of mechanical damage shall be cause for rejection.

4.6.4 Insert Retention

4.6.4.1 Procedure I. A pressure of 100±5 psig, or an equivalent mechanical load, shall be applied to the front face of the connector insert. The pressure shall be uniformly applied at a rate of 10-20 psi per second and held for 5-10 seconds at the specified value. Evidence of cracking, breaking, separation from the shell, or loosening of parts shall be cause for rejection.

4.6.4.2 Procedure II. The insert retention test shall be performed per 4.6.4.1, except that pressure shall be applied to both faces of the insert in turn.

4.6.5 Contact Retention

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4.6.5.1 Procedure I. The applicable forces listed shall be applied along the longitudinal axes of individual contacts in the firection tending to displace the contacts to the rear:

Contact	Load in Pounds +10%
22M	20
22	20
20	20
16	20
12	20

After preloading to 3 pounds, the force shall be applied at a rate of approximately 1 pound per second and maintained at full load for 5-10 seconds. No damage to contacts or insert shall result, nor shall the contacts be dislocated from their normal position in the connector more than 0.012 inch under the given load. Failure to meet these requirements shall be cause for rejection.

4.6.5.2 Procedure II. The contact retention test shall be performed per 4.6.5.1, followed by the same test applied in the opposite direction.

4.6.6 Coupling Pin Strength. One coupling pin in each receptacle shall be subjected to a 50 + 5 - 0 pound load applied to the swaged end along the major axis of the coupling pin and perpendicular to the pin centerline on the exposed portion of welded on brazed pins. A still test tip shall be used for application of the load on swaged pins. A steel test fixture contoured to the shape of the coupling pin shall be used for application of the load on the welded or brazed pins. Any displacement or perceptible loosening of the coupling pins resulting from the test shall be cause for rejection.

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4.6.7 Plating. The finish on coupling rings, hex, jam nuts, potting rings, shells, and contacts shall be checked for thickness by micro sectioning or other suitable means. Failure to meet these requirements shall be cause for rejection. This inspection may be performed on unassembled parts wherever a 3/4-inch diameter ball can contact the surface of the part.

4.6.8 Temperature Cycling.

4.6.8.1 Variety A. Mated connectors shall be subjected to the temperature cycling of MIL-STD-202, Method 102A. Test Condition C, except that steps 2 and 4 shall be of two minutes maximum duration and the temperatures of step 3 shall be 150° +5° -0°C (302°F) for Series I and IIA and 200° +5° -0°C (392°F) for Series IIB. Evidence of cracking, breaking, or loosening of parts shall be cause for rejection.

4.6.8.2 Variety B. Mated connectors shall be subjected to the temperature cycling test of MIL-STD-202, Method 102A, Condition D, except that steps 2 and 4 shall be of two minutes maximum duration and step 3 shall be 177° +5°, -0°C (351°F) for Series I and 150° +5°, -0°C (302°F) for Series II. Evidence of cracking, breaking, or loosening of parts shall be cause for rejection.

4.6.9 Engaging and Separating Forces. Cylindrical steel test pins, having spherical tips and surface finish not exceeding 3 microinches roughness, shall be engaged with and separated from the socket contacts. The forces necessary to insert the max. dia. pin and withdraw the minimum dia. pin shall be measured and shall meet the requirements listed following. The depth of insertion shall be not less than 0.205 inch measured from the face of the insert, or .140 measured from the end of the socket body, but the pins shall not bottom in the sockets.

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Test Pin Diameter in Inches			Forces in Ounces
Max. pin	Sizes 22 & 22M	0.0305 +.002 -.0000	12
	Size 20	0.041 +.0002 -.0000	18
	Size 16	0.0635 +.0002 -.0000	30 Variety A 24 Variety B
	Size 12	0.095 +.0002 -.0000	30 Variety A only
Min. Pin	Sizes 22 & 22M	0.0295 +.0000 -.0002	3/4
	Size 20	0.039 +.0000 -.0002	2
	Size 16	0.0615 +.0000 -.0002	2
	Size 12	0.093 +.0000 -.0002	3 Variety A only

Failure of the measured force to fall within these limits shall be cause for rejection

4.6.10 Test-Probe Damage

Step 1. All socket contacts shall be initially subjected to the contact engaging and separating force test. Contact voltage drop on each socket shall then be measured as follows:

Step 2. Apply 1 ampere and measure the millivolt drop across a mated pin/socket contact. Contact voltage drop shall be measured as shown in figure 5. Open circuit voltage shall be 6 volts direct current. Measurements shall be taken after the temperature has been stabilized.

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Step 3. The test-probe damage test shall begin with the socket contact being mounted into a socket-holding fixture. A test probe, in accordance with MS3321 shall be inserted. The weight of the test-probe shall be adjusted to produce a bending moment of 2 lb-in ±10 percent to the socket. The contact shall be rotated in one direction through 360 degrees so that a uniform force is applied to the inside surface of the socket.

Step 4. At the conclusion of the test probe damage test, step 2 shall be repeated, followed by step 1.

Failure to meet the requirements of this test shall be cause for rejection.

4.6.11 Dielectric Withstanding Voltage

Mated connectors connectors and unmated connector halves with pin contacts shall be placed in a suitable chamber capable of simulating altitudes of sea level, 50,000 feet, 70,000 feet, and 110,000 feet. The leads of all test circuits shall be brought out through the walls of the chamber. There shall be no wire splices inside the chamber. The wire ends of all leads shall be unsealed. The chamber shall be evacuated to each of the specified altitude pressure equivalents. The applicable test voltages shall be applied between three pairs of adjacent contacts and between three peripheral contacts and the shell in accordance with MIL-STD-202, Method 301. Leakage current shall not exceed .05 milliampere. Any evidence of dielectric breakdown or flashover shall be cause for rejection.

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Maximum Test Voltages, AC RMS, 60 CPS

	Service Rating M		Service Rating I		Service Rating II	
	Mated	Unmated	Mated	Unmated	Mated	Unmated
Sea Level	1,300	1,300	1,800	1,800	2,300	2,300
50,000 feet	800	550	1,000	600	1,000	800
70,000 feet	800	350	1,000	400	1,000	500
110,000 feet	800	200	1,000	200	1,000	200

4.6.12 Coupling Torque For qualification testing, mating halves shall be coupled and uncoupled, measuring the torques necessary. The torques required to couple and uncouple mating connector halves shall fall within the limits specified as follows:

Coupling Torques in Pound-Inches

Series I			Series II		
Shell Size	Max.	Min.	Shell Size	Max.	Min.
9	8	1	8	8	1
11	12	1	10	12	1
13	12	1	12	16	2
15	18	3	14	20	4
17	24	3	16	24	4
19	28	3	18	28	5
21	32	5	20	32	6
23	36	5	22	36	7
25	40	5	24	36	7

For acceptance testing, suitable gages may be used instead of the appropriate counterparts. A measured torque not within these limits shall be cause for rejection.

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4.6.13 Durability. Connector halves shall be mated and unmated 250 times at a rate not exceeding 200 cycles per hour. The test may be performed by hand or by mechanical means, but the coupling ring shall be operated as in normal service. Immediately after this test, the connectors shall be subjected to the Contact Resistance Test. Failure to satisfy the requirements of the Contact Resistance Test shall be considered as failure to pass the Durability Test and cause for rejection.

4.6.13.1 Contact Durability

The connectors shall be subjected to mating and unmating in a manner similar to service operation at a cycling rate of 4 per minute maximum for a total of 500 cycles. No mechanical assist is permitted during mating/unmating cycling. After every 25 mating/unmating cycles, two pin contacts shall be removed from the connector which are to be subjected to sodium polysulfide solution. Then immerse the two contacts in the polysulfide solution for 60 seconds at room temperature. The contacts shall then be removed from the solution and maintained at room temperature for 5 minutes. The contacts shall then be examined under 10X magnification. Exposed base metal will show a discoloration as a dark brown or black residue.

Continue removing two contacts and subjecting to polysulfide solution after every 25 mating/unmating cycles until all pin contacts have been removed from the connector half.

Any discoloration of pin contacts within the first 200 mating/unmating cycles shall be reason for rejection.

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The polysulfide solution shall be prepared as follows:

Sodium sulfide shall be dissolved in water until saturation occurs. The solution shall then be heated until all sodium sulfide has dissolved. The solution shall then be cooled to 25° ± 4°C. A minimum of 250 grams of flowers of sulfide shall then be added for every liter of solution. After 24 hours, the solution shall be further diluted with water to a specific gravity of 1.142. During test, the solution shall periodically be tested to determine that the solution has sufficient strength to blacken thoroughly a piece of clean untinned copper wire in 5 seconds. The test solution shall not be considered to be exhausted until it fails to blacken a piece of copper, as described previously.

4.6.14 Contact Resistance. Measurements, as shown in figure 5, shall be made across two mated pins and sockets assembled in the mated connector. Test shall be performed on mated pins and mated connectors only. Measurements shall be made from the plug end of the connector only. The test shall be conducted in the following manner: one ampere shall be applied and the millivolt drop across the two mated contact pins shall be measured after the temperature of the voltage of the power source shall not exceed 6 volts direct current. A measured potential drop exceeding the value specified shall be cause for rejection.

4.6.15 Insulation Resistance

4.6.15.1 Variety A

4.6.15.1.1 Insulation Resistance at Room Temperature. An insulation resistance test shall be performed on mated and unmated connectors in accordance with MIL-STS-202, Method 302, Test Condition B. Measurement

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shall be made between three pairs of adjacent contacts and between three peripheral contacts and the shell. Failure to meet the minimum requirement of 100,000 megohms for Classes E, P, and T, and 5,000 megohms for Class H shall be cause for rejection.

For lot acceptance testing, where it is undesirable to install actual contacts in connectors, simulated contacts and special techniques may be used in performing this test.

4.6.15.1.2 Insulation Resistance at Elevated Temperature. The insulation resistance test shall be repeated except that the connectors shall have been exposed to an ambient temperature as given below for a period of 30 minutes. Measurement shall be made while the connectors are still in the chamber at the specified temperature:

Series I and IIA	150 +5 -0°C (302°F)
Series IIB	250 +5 -0°C (392°F)

Failure to meet the following minimum requirements shall be cause for rejection.

	Series I & IIA	Series IIB
Classes E, P, and T	1,000 megohms	200 megohms
Class H	100 megohms	10 megohms

4.6.15.2 Variety B - An insulation resistance test shall be performed on mated and unmated connectors in accordance with MIL-STD-202, Method 302, Test Condition B. The voltage shall be applied between three pairs of adjacent contacts and between three peripheral contacts and the shell. Failure to meet the minimum requirement of 100,000 megohms for Classes T and P, and 5,000 megohms for Class H shall be cause for rejection.

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4.6.16 Sinusoidal Vibration. - Wired, mated connectors shall be subjected to the vibration test of MIL-STD-202, Method 204A, Test Condition D, with the following exceptions:

a. The specimens shall be subjected to a simple harmonic having an amplitude of:

50 - 2,000 Hz	40 g's ±10% or 0.06 inch double amplitude, whichever is less
2,000 - 3,000 Hz	20 g's ±10%

b. The entire frequency ranges of 50 to 2,000 Hz and 2,000 - 3,000 Hz shall be transversed at a sweep rate of 2 octaves per minute. Upsweep only is required.

c. Sinusoidal vibration shall be applied in each of two mutually perpendicular planes, one perpendicular and the other parallel to the longitudinal axis of the connector.

Receptacles shall be mounted on the vibration fixture by normal means. All contacts shall be wired in series circuit and 100-150 milliamperes of current shall be allowed to flow through the series circuit during vibration. Suitable means shall be employed to monitor the current flow and to indicate any discontinuity of more than 1 microsecond. The wire bundle shall be clamped to nonvibrating points at least 8 inches from the rear of the connector. Current discontinuity of 1 microsecond or more, disengagement of the mated connectors, evidence of cracking, breaking, or loosening of parts shall be cause for rejection.

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4.6.17 Shock. Wired, mated connectors shall be subjected to two shocks in each of two mutually perpendicular planes, one perpendicular and the other parallel to the longitudinal axis of the mated connector. Connectors shall be tested in accordance with Method 213 of MIL-STD-202C, Test Condition F.

Receptacles shall be mounted on a shock fixture by normal means. All contacts shall be wired in a series circuit and 100-150 ma. of current shall flow through the series circuit during shock. Suitable means shall be employed to monitor the current flow and to indicate any discontinuity of more than 1 microsecond. The wire bundle shall be clamped to fixed points at least 8 inches from the rear of the connector. Current discontinuity of 1 microsecond or more, disengagement of the mated connectors, evidence of cracking, breaking, or loosening of parts shall be cause for rejection.

4.6.18 Moisture Resistance. Wired, mated connectors shall be subjected to the moisture resistance test of MIL-STD-202, Method 106, except that steps 7a and 7b shall be omitted. Upon completion of step 6 of the final cycle, and while the connectors are still subjected to high humidity, the insulation resistance of mated Variety A crimp-type contact connectors shall be measured. Class H Variety A, and all Classes, Variety B connectors shall be removed and all external surfaces dried in a forced air oven at 66±5°C for 15 minutes, after which the insulation resistance of the mated connectors shall be measured. Insulation resistance values less than 100 megohms shall be cause for rejection.

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4.6.19 Thermal Shock (Class H Only) - (Variety A; Class H only, Variety B) Connectors shall be tested in accordance with Method 107B of MIL-STD-202C. The following details and exceptions apply:

a. Preparation: connectors shall be mated

b. Test Condition: A

c. High Temperature extremes: +150° +3°, 0°C

d. Exposure time for steps 1 and 3: 30 +5, -0 minutes

e. Maximum time for steps 2 and 4: 2 minutes

f. Examination after test: connectors shall be examined for evidence of mechanical damage.

4.6.20 Corrosion. Unmated connectors and individual contact samples shall be subjected to the salt spray test of MIL-STD-202, Method 101, Test Condition B (tin plated, Class H, Variety A) receptacles - 24 hours. Immediately after exposure, the surfaces of the specimens shall be thoroughly washed in tap water and dried in a circulatory oven at a temperature of 38 ±3°C (100°F) for a period of approximately 12 hours. Any exposure of base metal as a result of this test shall be cause for rejection.

4.6.21 Air Leakage

4.6.21.1 Variety A (Class H). The connector shall be mounted in a suitable test apparatus. A pressure differential of one atmosphere shall be applied across the connector. A suitable means to determine the leakage through the connector of air or other pressurizing gas, containing not less than 10

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percent helium by volume, shall be employed while the specified pressure is applied. Any evidence of a leakage in excess of 0.01 micron cubic ft/hr (1×10^{-1} cc/sec) during the test shall be cause for rejection.

4.6.21.2 Variety B

4.6.21.2.1 Class T (all) and Class P. The connector shall be mounted in a suitable test apparatus. The insert rear face (solder cup end) shall be completely covered with liquid. Air pressure of 10 to 15 psig shall be applied to the mating face of the connector insert and maintained for 10 to 15 seconds. During this period, the specimen under test shall be observed for escapement of air in the form of bubbles through any part of the insert or past the joints between insert and contacts, or insert and shell. The connector shall not be immersed in liquid for this test. Any evidence of air leakage during 10 to 15-second application shall be cause for rejection.

4.6.21.2.2 Class H. The connector shall be mounted in a suitable test apparatus. A pressure differential of one atmosphere shall be applied across the connector. A suitable means to determine the leakage through the connector of air or other pressurized gas, containing not less than 10 percent helium by volume, shall be employed while the specified pressure is applied. The air leakage rate shall be determined. Air leakage rate of more than 0.1 micron cubic foot per hour (1×10^{6} cc/sec) shall be cause for rejection.

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4.6.22 Altitude Immersion (Variety A Only)

Mated connectors shall be placed in a container of water at approximately 20°C and placed in an altitude chamber. All wire ends shall be located within the chamber and exposed to the chamber atmosphere, but not submerged. The exposed wire ends shall not be sealed. A quantity of salt, 5 percent by weight, shall be added to make the water conductive. The chamber pressure shall then be reduced to approximately one inch of mercury and maintained for thirty minutes. The chamber pressure shall then be slowly returned to atmospheric pressure. This shall be considered one cycle. Two additional cycles shall be performed. At the end of the last cycle, while the mated connectors are still submerged, the Insulation Resistance Test (room temperature) and the Dielectric Withstanding Voltage Test (sea level voltages) shall be performed upon the same circuits. Failure to meet an insulation resistance minimum of 2,000 megohms,or any evidence of dielectric breakdown or flashover shall be cause for rejection.

4.6.23 Solvent Immersion. Unmated connectors shall be immersed fully in the applicable fluid specified below for 20 hours. After removal from the fluid, each connector shall remain for one hour in free air at room temperature.

a. Jet fuel JP-4 to MIL-J-5624

b. Aircraft lubricating oil to MIL-L-9236

c. Hydraulic fluid to MIL-H-5606. Only new and uncontaminated hydraulic fluid shall be used.

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Following exposure to the above conditioning, connectors shall satisfactorily pass the next test in the sequence.

4.6.24 Temperature Durability. Wired, mated connectors shall be subjected to the indicated ambient temperature for a period of 1,000 hours.

Variety A

Series I and IIA---------- $150^{+3^{\circ}}_{-0^{\circ}}$C (302°F)

Series IIB---------- $200^{+3^{\circ}}_{-0^{\circ}}$C (392°F)

Variety B

Series I---------- $177^{+3^{\circ}}_{-0^{\circ}}$C (351°F)

Series II---------- $150^{+3^{\circ}}_{-0^{\circ}}$C (302°F)

4.6.25 Acceleration. Connectors shall be tested in accordance with Method 212 of MIL-STD-202C, Test Condition B, with the exceptions stated. Mated connectors shall be mounted on the acceleration test fixture by normal means. Connectors shall be wired as shown in figure 5. A g-level of 200 g's shall be applied in two opposite directions in each of three mutually perpendicular planes, one parallel to the longitudinal axis of the connector, and the other perpendicular to the longitudinal axis of the connector and perpendicular to each other. The test duration in each direction shall be two minutes in each direction, following centrifuge stabilization. Acceleration from 0 to 200 g's shall be accomplished within one minute. A current of 100-150 milliamperes shall be allowed to flow through the connector

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before, during, and at the conclusion of each test. Current discontinuity of 1 microsecond or more, disengagement of the mated connectors, evidence of cracking, breaking, or loosening of parts shall be cause for rejection.

4.6.26 Random Vibration. Mated connectors shall be mounted on the random vibration test fixture by normal means. Connectors shall be wired as shown in figure 5. Connectors shall be tested in accordance with the random vibration test, Method 214, MIL-STD-202C, Test Condition I, Test Letter J, with the following exceptions:

a. 100 Hz up to 3,000 Hz with power spectral density of 1 g^2/Hz, attenuated at not less than 12 decibels (db) per octave below 50 Hz and above 3,000 Hz.

b. 54.3 g RMS + 15 percent-5 percent overall.

The duration of the test shall be three minutes in each of two mutually perpendicular planes, one perpendicular and the other parallel to the longitudinal axis of the connector. A current discontinuity which exceeds 1 microsecond or any evidence of mechanical damage shall be reason for rejection.

4.6.27 Acoustical Noise

A reverberation-type test chamber, suitably formed and proportioned to produce, as close as possible, a diffuse sound field, the sound energy density of which is very nearly uniform throughout the enclosure. A pentagonal chamber configuration is recommended. Acute angles of adjacent chamber walls shall be avoided wherever possible. Acoustical generation

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reproduction and measuring equipment suitable to accomplish these tests shall be used.

Mated connectors shall be mounted on the acoustical noise test fixture by normal means. Connectors shall be wired as shown in figure 5. This test item shall be suspended in the test chamber by means of soft suspension cords such as soft springs or elastic cord. The natural frequency of all modes of suspension shall be less than 25 Hz. The test item shall be exposed on every surface to the sound field by centrally locating it in the test chamber. The test item volume should be no more than 10 percent of the test chamber volume. When the test chamber is a rectanguloid, no major surface of the test item shall be installed parallel to a chamber wall.

Step 1. - The sound pressure field shall be measured without the test item mounted in the test chamber. Measurements shall be made by using a microphone (more than one, if desired) to define the sound field within the test volume (central 10 percent of the chamber volume). The acoustics input of 155 db over-all sound pressure level (Ref. 2×10^{4} dynes/cm^2) shall then be introduced into the test chamber and adjusted to conform with the octave band spectrum specified in figure 6. The average sound pressure distribution (overall level) should be uniform within -2 to +4 db of the desired volume.

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Step 2 - The test item shall be placed in the chamber as specified. At least three microphones shall be monitored. They shall be located in proximity to each major dissimilar test item surface, at least 18 inches from the test item surface, or one-half the distance to the nearest chamber wall, whichever is less. The average over-all sound pressure distribution around the test item should be uniform within -2 to +4 db of the desired value.

Duration of the test will be 2 minutes. A current discontinuity which exceeds 1 microsecond or any evidence of mechanical damage shall be reason for rejection.

4.7 Test Methods, Variety A (Individual Crimp Contacts)

4.7.1 Examination of Contact. Contacts shall be carefully examined to determine conformance with this and applicable portions of referenced specifications or drawings which are not covered by other specific tests herein. Examination of the contact shall include inspection to determine if the required identification marking is present, if the contact is free of mechanical defects, and cracks around the external crimp area, and if the contact meets the physical requirements herein. The contacts shall be inspected through a device having a magnification power of approximately 3 diameters.

4.7.2 Crimp Deformation. Crimped contacts shall comply with the following requirements. Failure to do so shall be cause for rejection.

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The deformation of the contact, due to axial bending and out-of-roundness resulting from attachment to its wire with the crimping tool, shall not exceed 0.015 inch TIR which includes the maximum 0.0004 inch TIR permitted the contact during its manufacture. A typical method of determining the axial deformation is by the use of a ground precision check and accurate dial gage indicator equipment. The contact shall be checked on the "B" diameter for pin contacts, and the "C" diameter for socket contacts, both adjacent to the retention shoulder (figure 1). The probe of the dial indicator shall be positioned 1/32 inch from the wire entrance of the contact.

4.7.3 Temperature Durability. Mated, wired contacts shall be subjected to ambient temperatures of 150 + 3 -0°C (302°F) for Series I and IIA and 200 + 3 -0°C (392°F) for Series IIB, for a minimum of 1,000 hours. Following this conditioning, the mated contacts shall be allowed to return to room temperature prior to performing the subsequent tests. Following exposure to the previously mentioned conditioning, contacts shall satisfactorily pass the next test in the sequence.

4.7.4 Durability Conditioning. Mated, wired contacts shall be subjected to 250 cycles of engagement and separation at a rate not to exceed 200 cycles per hour. Engagement and separation shall be similar to that encountered in service. Failure to complete this test because of mechanical malfunction shall be cause for rejection.

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4.7.5 Tensile Strength. Contacts shall be placed in a suitable tensile testing machine and sufficient axial load applied to pull the wire out of the contact or break the wire. The wire shall not pull out nor shall the wire or contact break or become distorted to the extent of being unfit for further use before the minimum tensile strength specified in the following table is reached. The rate of travel of the head of the tensile machine shall be approximately 1 inch per minute. Wire breakage, not due to crimping, at less than the tensile loads shown as follows shall not constitute a failure.

Wire Size	Tensile Strength (lb min.)
28	3
26	5
24	8
22	12
20	20
18	40
16	50
14	70
12	110

Failure to comply with these requirements shall be cause for rejection.

4.8 Reliability Life Testing

4.8.1 Life Test Samples. Samples for life tests shall be chosen from the normal production run. The most dense insert arrangement of each service rating in the largest shell size for which qualification has been granted shall be represented in the lot. Suitable mating counterparts shall be provided. Manufacturers not producing mating connectors shall submit data substantiating that the tests were performed with approved counterpart connectors. In no case shall the number of sample units be less than ten.

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4.8.2 Test Preparation. Prior to preparation for life test, each connector shall be inspected for conformance to drawings and all standards and specifications referenced herein.

4.8.3 Physical Preparation.

4.8.3.1 Variety A. All sample connectors shall be wired with extruded TFE covered wire of nominal gage, conforming to MIL-W-16878/4 or MIL-W-22759.

4.8.3.2 Variety B.

a. All sample connectors shall be wired with extruded TFE covered wire of nominal gage conforming to MIL-W-16878/4 or MIL-W-22759.

b. TFE insulation shall be sodium etched.

c. After wires have been soldered, the rear face of connectors and inserts shall be cleaned with solvent and primed with Dow-Corning primer A409A, or equivalent.

d. All sample connectors shall be potted with Dow-Corning RTV silicone potting material, or equivalent.

4.8.4 Proof Tests

4.8.4.1 Pretests. Prior to beginning the Life Test, the following tests shall be performed.

	Test Paragraph
a. Contact Resistance	4.6.13
b. Insulation Resistance	4.6.14
c. Dielectric Withstanding Voltage	4.6.10

d. Mating and Unmating. Each assembly shall be subjected to 10 mating and unmating operations.

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4.8.4.2 Check Tests. At the end of each 300 unit cycles of the Life Test, the following tests shall be performed.

	Test Paragraph
a. Contact Resistance	4.6.13
b. Insulation Resistance	4.6.14
c. Dielectric Withstanding Voltage	4.6.10

4.8.4.2.1 Check Testing. Check tests shall be performed as specified in the applicable test paragraph except as follows:

a. Contact resistance. The maximum millivolt drop values shall be increased 100 percent.

b. Dielectric Withstanding Voltage. The applied voltages shall be:

(1) 400 vac rms for Service Rating M

(2) 600 vac rms for Service Rating I

(3) 900 vac rms for Service Rating II

c. Insulation resistance. 100 megohms minimum

4.8.5 Life Test Cycle. One cycle of the Life Test shall consist of the following:

a. Condition for a minimum of 30 minutes at standard test conditions.

b. Condition for 30 minutes at a temperature of:

Variety A: 150°C (302°F) Series I and Series IIA

200°C (392°F) Series IIB

Variety B: 177°C (315°F) Series I

150°C (302°F) Series II

All tolerances +5 -0°C.

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c. Condition for a minimum of 30 minutes at standard test conditions.

d. Condition for 30 minutes at a temperature of -55°C (-67°F).

e. Condition for a minimum of 30 minutes at standard test conditions.

f. One cycle of unmating and remating.

g. Vibration test as described in 4.6.16 except that the time shall be limited to one hour per cycle (note 3 and 4.8.6).

h. Humidity Test of MIL-STD-202, Method 103, exposure only.

EXCEPTION: Exposure time shall be 4 hours.

NOTES:

1. Unless otherwise specified, all times for the above testing shall be within ±5 minutes.

2. Standard test conditions are defined as:

a. Temperature from 20°C (68°F) to 30°C (86°F).

b. Relative humidity from 30 to 80 percent.

c. Barometric pressure 24 to 31 inches Hg.

3. For each successive cycle of vibration, the direction shall be changed so that vibration in both directions along the longitudinal and transverse axes will be accomplished.

4.8.6 Failure Criteria. Failures to pass the Check Tests, electrical discontinuities in excess of 10 microseconds duration during vibration, loosening or separation of parts, or seizing of parts so as to prevent mating or unmating by hand shall constitute failure to pass the Life Test. More than one failure in 1,500 unit cycles shall constitute failure to demonstrate the required reliability at Reliability Level 2.

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4.8.7 Reliability Level Documentation. The demonstrated reliability level shall be fully documented and records of all testing shall be made available for inspection by a duly authorized representative of the procuring activity on request.

4.8.8 Reliability Indication. The purpose of the Reliability Life Test is to demonstrate an initial mean cycle between failure (MCBF) of not less than 500 cycles with 90-percent confidence. Successful completion of reliability life testing on representative samples as defined in 4.7.1 shall establish the reliability level for all qualified connectors. Subsequent test cycles and failures may be accumulated on a continuing basis, and the manufacturer's reliability level shall be in accordance with the following table:

MCBF with Confidence Level		Reliability Indicator
500	90%	1
1,000	90%	2
1,500	90%	3
2,000	90%	4
		5

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Unless otherwise specified, each connector, prtected by dust covers, shall be packed and sealed in a bag conforming to MIL-B-22020 or MIL-B-22205. If the MIL-B-22205 bag is used, tarnish inhibiting paper shall be placed in the bag in such a manner that identification markings on the connectors will not be obscured. The tarnish inhibiting paper

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shall be a quality equal to, or better than, "Silver Saver Paper", available from Orchard Paper Company, St. Louis, Missouri. If metal protection caps or other accessory hardware are required to be shipped with the connector, contracting officer should so specify.

5.2 Packing, All Levels. A number of connectors, specified by the procuring activity, preserved and packaged in accordance with 5.1, shall be protected in accordance with MIL-STD-129. The lot number (paragraph 3.8) shall be included in the marking.

5.3.1 Marking Exterior Containers. Exterior containers shall be marked in accordance with MIL-STD-129. The lot number (paragraph 3.8) shall be included in the marking.

5.3.1.1 Nomenclature. Shipment nomenclature shall be as specified in the contract.

6. NOTES

6.1 Intended Use. The various configurations of the connectors are intended for use as follows:

6.1.1 Variety A

a. Class E, Environment Resisting. Provisions are made for sealing around wires on rear end.

b. Class P, intended for use in applications where sealing around wires is accomplished by potting.

c. Class T, general duty, is supplied without rear accessories to enable the user to install accessories of his choice.

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d. Class H, intended for use in applications wherein pressure must be contained by the connectors across the walls or panels on which they are mounted.

6.1.2 Variety B

a. Class T, general duty, no provisions are made for sealing around wires on rear end.

b. Class P, intended for use in applications where sealing around wires is accomplished by potting.

c. Class H, intended for use in applications wherein pressure must be contained by the connectors across the walls or panels on which they are mounted.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Nomenclature by Variety, Class, Type, Finish, Style, Size, and Government Part Number, plus Kind.

c. Method of preparation for delivery.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and the manufacturers are urged to arrange to have the products they proposed to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products Lists

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is, Picatinny Arsenal, ATTN: SMUPA-TN4, Dover, New Jersey 07801, however, information pertaining to qualification of products may be obrained from the Defense Electronic Supply Center, ATTN: DESC-EMM, 1507 Wilmington Pike, Dayton, Ohio 45420: Phone - Area Code 513, 252-6551, extension 29234.

Preparing Activity
Army-MU

Custodian:
ARMY-MU

Project Number 5935-A675

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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